Four-zero tail-palindromic polynomials over $\mathbb{F}_2$: reciprocity and transformation boundaries

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Abstract

We study monic even-degree polynomials over $\mathbb{F}_2$ whose nonleading coefficients form a palindrome with constant term one and exactly four zeros. We classify their intersections with full-degree quadratic rational outputs and exclude iterated doubling from an even input degree. An exact reciprocal common divisor gives an arithmetic screen and a closed count of surviving position pairs. For the polynomial obtained by cubing all roots, counted with multiplicity, we prove that the tail is never palindromic. The corresponding quintic image is tail-palindromic exactly for one degree-10 exception or an explicit arithmetic-progression family; its number of missing tail coefficients is always 2 modulo 4. Sparse norm identities and independently checkable finite parameter certificates prove these root-power classifications for all degrees. They exclude degree-preserving cube- and fifth-root extraction within the four-zero family. Reproducible computations identify 2663 primitive polynomials at the 98 even degrees from 6 to 200 and verify first witnesses through degree 300. The primitive-existence conjecture C_4 remains open.

1 The family and the question

Rational pullbacks and powers of roots give different transformations of a polynomial. Quadratic rational transformations are a standard tool for constructing irreducible polynomials of doubled degree [8]; power substitution has classical irreducibility criteria [11, Section 2]. This paper determines when such constructions are compatible with a rigid coefficient constraint: a palindromic tail with exactly four zero coefficients.

Put $n = 2k \geq 6$ and $S_n(x) = 1 + x + \cdots + x^n$. We consider

$$f_{n,i,j}(x) = S_n(x) + x^i + x^j + x^{n-1-i} + x^{n-1-j}, \qquad 1 \leq i < j \leq k-1, \tag{1}$$

over $\mathbb{F}_2$. These are exactly the monic degree-n polynomials with constant term one whose nonleading coefficient vector is a palindrome with four zero entries. The restrictions on i, j exclude the constant position. There are $\binom{k-1}{2}$ candidates. Let $N_4(n)$ denote the number of primitive polynomials in this family. Here primitive means irreducible with a root of multiplicative order $2^n - 1$; standard background is in [2].

Conjecture 1.1 (C_4)**.** For every even $n \geq 6$, one has $N_4(n) > 0$.

Primitive polynomials with prescribed coefficients are studied systematically by Cohen [1]. In the present family the field is fixed, all but four tail coefficients are one, and the four missing positions are coupled by reversal. The number of coefficient constraints therefore grows with the degree. A tail palindrome can also be encoded by the self-reciprocal polynomial $xf(x) + 1$ of odd degree $n + 1$. This describes its support; irreducibility and primitivity refer throughout to f itself.

Theorems 3.3 and 3.8 completely classify the full-degree four-zero quadratic outputs in the translated Q and translation-invariant branches. A trace obstruction yields uniform reducibility when the number of missing tail coefficients is divisible by four (Proposition 3.7). Corollary 3.9 combines the branches and rules out iterated doubling from an even input degree. These proofs convert the coefficient constraint into coupled translation identities and solve them by binary submask arguments.

Section 2 gives a complementary arithmetic screen: an exact reciprocal common divisor and a closed count of the surviving position pairs (Propositions 2.2 and 2.4).

Section 4 studies the degree-preserving operation of raising every root to the third or fifth power, with multiplicities retained. Theorem 4.1 gives a cubic obstruction and a complete quintic tail-symmetry classification. The quintic images have 2 modulo 4 missing coefficients, so neither operation stays within the four-zero family. The proof uses a fixed sparse feedback polynomial and exact finite parameter certificates, distinct from the bounded-degree arithmetic enumeration.

Section 5 records primitive counts and first witnesses, and Section 6 states the remaining existence questions. Appendix A gives classical coefficient and finite-field descriptions. Appendix B specifies the exhaustive root-power certificate checks and their reproducibility.

2 The reciprocal obstruction

Tail symmetry turns polynomial reciprocity into an explicit common-factor test. In the four-zero family, the forced factor is determined by two integer gcds, and the number of surviving position pairs has a closed formula. The calculation uses the binomial gcd identity [2] and unit counts expressed by the second Schemmel function [5, Section 1].

Lemma 2.1 (Reciprocity and multiplicities)**.** *Let $f = x^n + \sum_{r=0}^{n-1} c_r x^r$, where n is even, $c_0 = 1$, and $c_r = c_{n-1-r}$. Write $f^*(x) = x^n f(1/x)$. Then*

$$f^* = xf + x^{n+1} + 1, \qquad \gcd(f, f^*) = \gcd(f, x^{n+1} + 1).$$

This gcd is squarefree. In particular, every self-reciprocal irreducible factor of f occurs with multiplicity one. If a non-self-reciprocal irreducible g and g^* both divide f, the smaller of their multiplicities is one.

Proof. The coefficient symmetry gives the displayed identity, hence the gcd identity by subtraction. The derivative of $x^{n+1} + 1$ is x^n, which is coprime to it; thus its divisors are squarefree. A self-reciprocal factor has the same multiplicity in f and f^*. For a reciprocal pair the two multiplicities are interchanged, so their minimum is the multiplicity in the gcd. This proves the claims. □

Proposition 2.2 (Exact reciprocal common divisor)**.** *For $f = f_{n,i,j}$ from (1), put*

$$N = n + 1, \quad \Delta = j - i, \quad R = n - 1 - i - j, \quad a = \gcd(N, \Delta), \quad b = \gcd(N, R), \quad h = \gcd(a, b).$$

Then, as an exact polynomial quotient in $\mathbb{F}_2[x]$,

$$\gcd(f, f^*) = \frac{(x^a + 1)(x^b + 1)}{(x + 1)(x^h + 1)}, \qquad \deg \gcd(f, f^*) = a + b - h - 1. \tag{2}$$

Consequently, irreducibility of f requires $a = b = 1$.

Proof. The nonleading terms of f occur in pairs, so $f(1) = 1$. By Lemma 2.1, with $S_n = (x^N + 1)/(x + 1)$,

$$\gcd(f, f^*) = \gcd(f, S_n) = \gcd(S_n, P), \qquad P = x^i(1 + x^\Delta)(1 + x^R).$$

The oddness of N makes S_n squarefree. For positive u, v, the Euclidean algorithm on the exponents gives $\gcd(x^u + 1, x^v + 1) = x^{\gcd(u,v)} + 1$ over $\mathbb{F}_2$. Applying this identity yields

$$\gcd(S_n, 1 + x^\Delta) = \frac{x^a + 1}{x + 1}, \qquad \gcd(S_n, 1 + x^R) = \frac{x^b + 1}{x + 1}.$$

Since S_n is squarefree, its gcd with their product is their least common multiple. Their intersection is $(x^h + 1)/(x + 1)$, proving (2). If this gcd were all of f, monicity and equal degrees would give $f = f^*$, hence $(x + 1)f = x^N + 1$ and $f = S_n$, contrary to its four zero coefficients. Thus a nonconstant gcd is a proper divisor. The right side is constant exactly when $a = b = 1$. □

Corollary 2.3 (An explicit infinite reducible family)**.** *For every odd $m \geq 3$, the valid position pair $(n, i, j) = (3m - 1, m - 2, m)$ satisfies*

$$\begin{aligned} f_{3m-1,m-2,m}(x) = (1 + x + \cdots + x^{m-1}) \\ \cdot (x^{2m} + x^{m+1} + x^{m-1} + x^{m-2} + 1). \end{aligned} \tag{3}$$

In particular it is reducible, and the displayed forced divisor has unbounded degree $m - 1$.

Proof. The inequalities $1 \leq m - 2 < m \leq (3m - 3)/2$ make the positions valid. Here $N = 3m$, $\Delta = 2$, and $R = m$, so $a = 1$, $b = m$ in Proposition 2.2. For the explicit quotient, group the geometric sum as $S_{3m-1} = S_{m-1}(1 + x^m + x^{2m})$. The four deleted terms sum to

$$x^{m-2}(1 + x^2)(1 + x^m) = S_{m-1}x^{m-2}(x + 1)^3.$$

Adding these expressions and cancelling the two x^m terms inside the second factor proves (3). □

Proposition 2.4 (Number of positions surviving the screen)**.** *For odd $N = n + 1$, let φ be Euler's totient and put*

$$\mathfrak{S}_2(N) = N \prod_{p \mid N} \left(1 - \frac{2}{p}\right).$$

The number of four-zero position pairs with $\gcd(f, f^*) = 1$ is

$$E_N = \binom{\varphi(N)/2}{2} - \frac{\mathfrak{S}_2(N) - 1}{2}. \tag{4}$$

Thus $N_4(n) \leq E_{n+1}$. This is only an upper bound; $E_{n+1} > 0$ does not imply $N_4(n) > 0$.

Proof. Set $u = i + 1$, $v = j + 1$. The two coprimality conditions in Proposition 2.2 are precisely that $u - v$ and $u + v$ be units modulo N. Since 2 is invertible, the map $(u, v) \mapsto (u - v, u + v)$ is a bijection on $(\mathbb{Z}/N\mathbb{Z})^2$, giving $\varphi(N)^2$ ordered pairs before excluding zero coordinates. The cases $u = 0$ and $v = 0$ contribute $\varphi(N)$ each and are disjoint. The independent sign changes of u, v and their interchange act freely on the remaining pairs: admissibility excludes $u = \pm v$, and a nonzero coordinate cannot equal its negative. Dividing by eight gives $\varphi(N)(\varphi(N) - 2)/8$ unordered pairs of distinct nonzero sign classes.

Choose representatives $1, \ldots, k$ of those classes. The allowed positions use $2, \ldots, k$, so the class represented by 1 must be removed. The Chinese remainder theorem counts $\mathfrak{S}_2(N)$ residues v for which both $v - 1$ and $v + 1$ are units: modulo each prime $p \mid N$ two residues are forbidden. One residue is $v = 0$; the others form pairs $\{v, -v\}$. Hence the removed class has $(\mathfrak{S}_2(N) - 1)/2$ neighbors. Subtraction proves (4). □

Remark 2.5 (Factorization examples and fixed positions). For example,

$$f_{8,1,2} = (x^3 + x^2 + 1)(x^5 + x^2 + 1)$$

has trivial reciprocal gcd and is reducible. Repeated factors also occur:

$$f_{14,2,5} = (x^2 + x + 1)(x^6 + x^4 + x^3 + x + 1)^2.$$

The squarefreeness hypothesis in the classical Stickelberger–Swan factor-parity criterion [3] must therefore be checked. An odd number of factors still need not mean irreducibility: $f_{12,4,5}$ has three distinct factors of degrees $3, 4, 5$.

For fixed i, j, one also has

$$(x+1)f_{n,i,j} = x^{n-1-j}A_{i,j} + B_{i,j},$$

where $A_{i,j} = x^{j+2} + (x+1)(x^{j-i}+1)$ and $B_{i,j} = 1 + (x+1)(x^i + x^j)$ are fixed polynomials. This places fixed-position factor-parity questions in the classical framework of Fredricksen, Hales and Sweet [4].

3 Quadratic-transform intersections

3.1 The translated Q branch

For a degree-m polynomial h, define

$$\mathcal{T}(h)(x) = (x+1)^m h\left(\frac{x^2}{x+1}\right), \qquad \mathcal{M}_n(f)(x) = (x+1)^n f\left(\frac{x}{x+1}\right). \tag{5}$$

The map $\mathcal{T}$ is the classical Q-transform followed by the translation $x \mapsto x+1$; explicitly, $\mathcal{T}(h)(x) = h^Q(x+1)$. The equivalence between $\mathcal{M}_{2m}(f) = f$ and $f = \mathcal{T}(h)$ is a special case of Mattarei and Pizzato [8, Lemma 9]: take $a = b = 1, c = 0$, numerator x^2 and denominator $x+1$ in their notation. The nondegeneracy conditions hold, and the representation is unique; indeed the summands $x^{2j}(x+1)^{m-j}$ have distinct degrees $m + j$. More general invariant representations are given by Reis [12, Theorem 1.3]. We combine these invariant representations with the four-zero coefficient condition to determine every compatible output, including reducible polynomials.

Write $\Delta a = a(x+1) + a(x)$, and use reversal with respect to a specified formal degree when the leading coefficient may vanish.

Lemma 3.1 (A descent for tail palindromes)**.** *Let $f = x^{2k} + \sum_{j=0}^{2k-1} c_j x^j$, with $c_0 = 1$ and $c_j = c_{2k-1-j}$. Set $d = k-1$ and*

$$e = \sum_{a=0}^{k-1} c_{2a} x^a, \qquad o = x^d e(1/x).$$

Then

$$\mathcal{M}_{2k}(f) = f \quad \Longleftrightarrow \quad \Delta e = 0 \text{ and } e = \Delta(xo). \tag{6}$$

If the tail of f has $2s$ zeros, the formal coefficient vector of e has s zeros.

Proof. Tail symmetry gives $f = x^{2k} + e^2 + xo^2$. Put $p = xf + x^{2k+1} = xe^2 + x^2o^2$, and homogenize p to formal degree $2k+1$, obtaining $P(X,Y)$. The tail condition is exactly $P(X,Y) = P(Y,X)$. Moreover, $\mathcal{M}_{2k}(f) = f$ is equivalent to $P(X, X+Y) = P(X,Y)$. Conjugating this substitution by

the interchange of X, Y gives $(X, Y) \mapsto (X + Y, Y)$. Hence, for this symmetric P, the condition is equivalent to $\Delta p = 0$. Direct expansion gives

$$\Delta p = x(\Delta e)^2 + \big(e(x+1) + \Delta(xo)\big)^2.$$

The two summands have disjoint odd and even supports. Thus both vanish exactly under the conditions in (6). Each tail mirror pair contains one even and one odd position, which proves the assertion about zeros. □

Lemma 3.2 (Two missing coefficients and translation)**.** *Let d be positive and even, and $1 \le a < b \le d-1$. Then $e = S_d + x^a + x^b$ satisfies $\Delta e = 0$ exactly in the following cases:*

	d	$\{a, b\}$
I	$2^r - 2,\ r \ge 3$	$\{2^u, 2^v\},\quad 0 \le u < v \le r-1$
II	$2^r,\ r \ge 2$	$\{2^r - 1, 2^t\},\quad 0 \le t \le r-1$
III	$2^r + 2^s - 2,\ r > s \ge 2$	$\{2^r - 1, 2^s - 1\}.$

(7)

Proof. A binary submask of an integer is obtained by retaining any subset of its nonzero binary digits; it is proper when it differs from the integer. The binary expansion identity $(x+1)^v = \prod_{j:\, v_j = 1} (x^{2^j} + 1)$ shows that Δx^v, for $v > 0$, consists of the proper binary submasks of v, and has $2^{\mathrm{wt}_2(v)} - 1$ terms. Consequently $\Delta x^v = 1$ exactly for powers of two. For distinct positive v, w, equality $\Delta x^v = \Delta x^w$ also occurs exactly when both are powers of two: with at least two binary digits, the singleton submasks recover all set digits. We will also use

$$x(x+1)\Delta S_t = x^{t+2} + (x+1)^{t+2} + 1. \tag{8}$$

It follows that $\Delta S_t = 0$ exactly when $t + 2$ is a power of two.

For $0 \le b \le t$, the one-missing-coefficient polynomial $S_t + x^b$ is translation invariant exactly when

$$(t, b) = (2^s - 2, 0) \quad \text{or} \quad (2^s - 1, 2^s - 1), \qquad s \ge 1.$$

Indeed equality of its values at $0, 1$ forces $b = 0$ if t is even. If t is odd, translation invariance forces its leading term to cancel, since a nonzero invariant polynomial has even degree, so $b = t$. Equation (8) completes both cases, including the zero polynomial at $t = b = 0$.

Put $d = 2u$ and write $e = E^2 + xO^2$. Odd and even coefficient comparison gives $\Delta e = 0 \iff \Delta O = 0,\ \Delta E = O$. The useful residual is

$$B_u = S_{u-1} + \Delta S_u = \frac{x^{u+1} + (x+1)^{u+1} + 1}{x}.$$

If a, b are even, then $O = S_{u-1}$. Thus $u+1$ is a power of two and $B_u = 0$. The remaining condition is $\Delta(x^{a/2} + x^{b/2}) = 0$, so $a/2, b/2$ are powers of two. These are precisely the even-even pairs in type I.

If a, b are odd, then $E = S_u$ and $B_u = x^{(a-1)/2} + x^{(b-1)/2}$. Its two terms force $\mathrm{wt}_2(u+1) = 2$. Writing $u + 1 = 2^v + 2^w$, $v > w \ge 0$, gives $B_u = x^{2^v - 1} + x^{2^w - 1}$. This yields type III when $w > 0$; $w = 0$ yields type II with $t = 0$.

Finally suppose the missing positions are $2A$ and $2B + 1$. Then $O = S_{u-1} + x^B$ and $B_u = x^B + \Delta x^A$. The one-missing-coefficient classification gives either $u = 2^s - 1, B = 0$, or $u = 2^s, B = u - 1$. In the first case $B_u = 0$; in the second $B_u = x^{u-1} + 1$. Both force $\Delta x^A = 1$, so A is a power of two. They give respectively type I with one missing position 1, and type II with $t > 0$. These cases exhaust the parities. All displayed pairs satisfy the required equations on substitution, so the classification is sufficient as well as necessary. □

Theorem 3.3 (Complete four-zero intersection)**.** *For the candidates* (1)*,* $\mathcal{M}_n(f_{n,i,j}) = f_{n,i,j}$ *if and only if one of the following holds:*

$$n = 2^R + 2^S - 2, \quad R > S \geq 2, \qquad (i,j) = (2^S - 2, 2^S - 1); \tag{9}$$

$$n = 2^R + 2, \quad R \geq 3, \qquad (i,j) = (3, 2^{R-1}); \tag{10}$$

$$n = 14, \qquad (i,j) \in \{(2,4), (2,5), (4,5)\}. \tag{11}$$

Equivalently, these are exactly the four-zero polynomials in the image of $\mathcal{T}$*.*

Proof. We first use coefficient comparison to reduce the four-zero condition to the two-zero polynomial in Lemma 3.1. We then apply the three types in Lemma 3.2 and impose the second descent identity on each type.

Write $n = 2k$. The coefficient of x^{n-1} in $\mathcal{M}_n(f)$ is $f'(1) = k-2$ in $\mathbb{F}_2$, since the four missing tail positions include two odd positions: each pair $\{r, n-1-r\}$ contains one odd position because $n-1$ is odd. Its value in f is 1, so k is odd. Comparison of the x^2 coefficients gives $[x]f = \binom{2k}{2} = k = 1$. Thus positions $1, n-2$ are not missing. In Lemma 3.1, $e = S_d + x^a + x^b$ is therefore monic of even degree $d = k-1$, with $1 \leq a < b \leq d-1$.

By Lemma 3.2, e has one of the three types in (7). The second descent condition is equivalent to

$$x^a + x^b + \Delta x^{d+1-a} + \Delta x^{d+1-b} = \frac{x^{d+2} + (x+1)^{d+2} + 1}{x}. \tag{12}$$

For type I the right side is zero. The binary masks $d+1-a$ and $d+1-b$ have all r digits set except the distinct digits specified by a, b. Their proper-submask supports have symmetric difference of size $2^{r-1} - 2$: among all submasks the difference has size 2^{r-1}, and deleting the two top masks removes two terms. Equation (12) requires that difference to be the two terms $x^a + x^b$, so $r = 3$. For $r = 3$ it is exactly that pair, giving $d = 6$ and the three cases (11).

For type II put $\{a, b\} = \{d-1, 2^t\}$. The equation becomes

$$\Delta x^{d+1-2^t} = x^{2^t} + x + 1.$$

For $t = 0$ both sides are 1. For $t \geq 1$ the right side has three terms, while $d+1-2^t$ has $r-t+1$ set digits. Thus $2^{r-t+1} - 1 = 3$, forcing $t = r-1$, which also suffices. The two choices yield respectively (9) with $S = 2$, and (10).

For type III the two Δ terms are both 1 and cancel. The right side is $x^a + x^b$, so every such pair satisfies the condition. Restoring the even zero positions $2a, 2b$ and their tail mirrors gives (9) with $R = r+1, S = s+1 \geq 3$. All these steps are reversible, proving the assertion in every degree. The equivalent image statement follows from the classical representation preceding Lemma 3.1. □

Corollary 3.4 (Reducibility and failure of iteration)**.** *Every member of* (9) *is reducible. If* h *is itself a four-zero polynomial of the form* (1)*, then* $\mathcal{T}(h)$ *is never a member of that family. Among the three polynomials in* (11)*, only* $f_{14,2,4}$ *is irreducible.*

Proof. For $f = \mathcal{T}(h)$ let $m = n/2$; its constant term gives $h(0) = 1$, and the x^2 coefficient gives

$$[x]h = [x^2]f + \binom{m}{2}. \tag{13}$$

In (9), $S = 2$ gives $m \equiv 1 \pmod 4$ and $[x^2]f = 0$, while $S \geq 3$ gives $m \equiv 3 \pmod 4$ and $[x^2]f = 1$. Hence $[x]h = 0$ in either case. If h is reducible, multiplicativity of the transform makes f reducible.

If h is irreducible, the classical binary Q criterion [16, Theorem 1.1] says that h^Q, and hence $f = h^Q(x+1)$, is irreducible exactly when $[x]h = 1$. This proves the first assertion.

The theorem, or its coefficient proof, gives $n \equiv 2 \pmod 4$ for every four-zero output. A four-zero input has even degree, so its transformed degree is divisible by four. This proves the iteration obstruction for this particular transform.

For the exceptional pairs $(2,4), (2,5), (4,5)$ the corresponding h are, respectively,

$$x^7 + x^6 + x^3 + x + 1, \quad x^7 + x^6 + x^3 + x^2 + x + 1, \quad x^7 + x^6 + x^4 + x^2 + 1.$$

The first satisfies $x^{128} \equiv x \pmod h$ and $\gcd(h, x^2 + x) = 1$, so is irreducible by the prime-degree Rabin test; its linear coefficient is 1. The second has root 1, and the third has zero linear coefficient. The same Q criterion finishes the proof. □

Proposition 3.5 (Explicit descent for the B family)**.** *For $R \geq 3$, put $q = 2^{R-1}$. The unique polynomial with $\mathcal{T}(h_R) = f_{2q+2,3,q}$ is*

$$h_R(x) = 1 + x + x^2 + (1+x)\left(x^q + \sum_{t=2}^{R-2} x^{q-2^t+1}\right), \tag{14}$$

where the sum is empty for $R = 3$. Moreover, $\deg h_R = q + 1$ and $[x]h_R = 1$.

Proof. Set $L(x) = \sum_{a=0}^{R-2} x^{2^a}$ and

$$F(x) = x^{q+2} h_R(1/x) = (x^2 + x + 1)x^q + (x+1)(L(x) + x^2).$$

For $s = X^{-1}$ and $w = s + s^2$, telescoping gives $L(w) = s + s^q$; substitution and collection of terms give

$$s^{-2q-4} F(w) = (X+1) f_{2q+2,3,q}(X).$$

The left side is $(X+1)\mathcal{T}(h_R)(X)$, proving the formula; the distinct leading degrees of the transform basis give uniqueness. The exponents in the sum lie between $q/2 + 1$ and $q - 3$, so the degree and linear coefficient follow directly. □

The classical Q criterion makes irreducibility of h_R equivalent to that of its output.

3.2 The translation branch

We next impose $f(x+1) = f(x)$. The invariant representation $f = H(x^2+x)$ and its Artin–Schreier irreducibility criterion are classical [15, Lemma 2.1 and Theorem 2.2]. The additional constraints below come from the tail palindrome and its missing coefficients.

Lemma 3.6 (Degree compatibility)**.** *If a monic tail-palindromic polynomial f of even degree $n \geq 6$ satisfies $f(x+1) = f(x)$, then $n + 2$ is a power of two.*

Proof. Let $F(X,Y) = Y^n f(X/Y)$ and $N = n + 1$. Tail symmetry gives

$$XF(X,Y) + YF(Y,X) = X^N + Y^N.$$

By translation invariance, put $A = F(X,Y) = F(X+Y,Y)$, $B = F(Y,X) = F(X+Y,X)$ and $C = F(X,X+Y) = F(Y,X+Y)$. Applying the displayed identity to three pairs of variables yields

$$\begin{aligned} XA + YB &= X^N + Y^N, \\ (X+Y)A + YC &= (X+Y)^N + Y^N, \\ XC + (X+Y)B &= X^N + (X+Y)^N. \end{aligned}$$

Multiply these equations by $X+Y$, X and Y, respectively, and add. All terms on the left cancel, leaving $(X+Y)^{n+2} = X^{n+2} + Y^{n+2}$. In characteristic two this forces $n+2$ to be a power of two: if its binary expansion has at least two nonzero digits, a singleton proper submask gives a nonzero mixed binomial term. □

Proposition 3.7 (A missing-coefficient obstruction)**.** *Let f be monic of even degree $n = 2k \geq 6$, with constant term one and a tail palindrome containing $2s$ zeros. If $f(x+1) = f(x)$, write $f = H(x^2+x)$ with H monic of degree k. Then*

$$[x^{k-1}]H = s \pmod 2. \tag{15}$$

In particular, if the number of tail zeros is divisible by four, then f is reducible.

Proof. With e, o as in Lemma 3.1, tail symmetry gives $f = (x^k+e)^2 + xo^2$. Odd and even coefficient comparison gives

$$\Delta f = 0 \quad \Longleftrightarrow \quad \Delta o = 0, \qquad \Delta(x^k+e) = o. \tag{16}$$

Indeed $\Delta f = x(\Delta o)^2 + [\Delta(x^k+e) + o(x+1)]^2$. The formal coefficient vector of o has length k and s zeros, so $[x^{n-2}]f = o(0) = o(1) = k-s$ in $\mathbb{F}_2$. Lemma 3.6 gives $k = 2^{R-1}-1 \equiv 3 \pmod 4$, where $R \geq 3$. The classical coefficient comparison in $H(x^2+x)$ therefore gives

$$[x^{k-1}]H = [x^{2k-2}]f + \binom{k}{2} = s \pmod 2.$$

Now suppose that s is even. If H is reducible, so is its composition. Otherwise let β be a root of the irreducible H in $\mathbb{F}_{2^k}$. Its absolute trace is $[x^{k-1}]H = 0$, so $z^2+z = \beta$ has a solution in that same field. Thus f has a root of degree at most k and is reducible. The last step is classical: specialize [14, Theorem 2.2(2)] to $p=2$, $n=r=1$ and $a=1$ in Agou's notation, with irreducible outer polynomial H; see also [15, Theorem 2.2]. The same binary specialization follows from Long [13, Theorems 4.1–4.2, pp. 69 and 71]; the basic Artin–Schreier trace criterion is classical, as noted in [13, p. 63]. The associated high-coefficient restriction appears in [15, Theorem 2.8(3)]. □

The restriction $n \geq 6$ is essential: x^2+x+1 is a translation-invariant irreducible tail palindrome with no zeros. For odd s, (15) only removes this trace obstruction; it does not establish irreducibility of H.

Theorem 3.8 (Complete translation intersection)**.** *The polynomial $f_{n,i,j}$ in* (1) *satisfies $f_{n,i,j}(x+1) = f_{n,i,j}(x)$ if and only if*

$$(n,i,j) \in \{(14,3,4), (30,5,9), (30,5,12), (30,9,12)\}. \tag{17}$$

All four polynomials are reducible.

Proof. In (16), put $d = k-1$. The polynomial o is monic of degree d and has two missing coefficients. Since k is odd, $o(1) = k-2 = 1$; invariance gives $o(0) = 1$. Thus $o = S_d + x^a + x^b$ with $1 \leq a < b \leq d-1$, and $e = S_d + x^{d-a} + x^{d-b}$. By Lemma 3.6, $d = 2^r - 2$, $r \geq 2$. The case $r = 2$ has only one interior position and is impossible. For $r \geq 3$, Equation (8) gives $\Delta S_d = 0$. Hence $\Delta x^a = \Delta x^b$, which, by the binary-support observation in the proof of Lemma 3.2, forces a, b to be distinct powers of two. The second condition in (16), using $\Delta x^{d+1} = S_d$, becomes

$$\Delta x^{d-a} + \Delta x^{d-b} = x^a + x^b. \tag{18}$$

If a, b are both even, take square roots in this equation. The two exponents on the left are $(r-1)$-digit all-one binary masks with different singleton digits removed. Their full submask supports have symmetric difference of size 2^{r-2}; removing the two top masks leaves $2^{r-2}-2$ terms. The right side has two terms, so $r=4$. For three-digit masks the remaining difference is precisely the two removed singleton digits; thus all pairs from $\{2,4,8\}$ satisfy (18).

Otherwise $a=1$ and b is an even power of two. The term Δx^{d-b} has only even exponents. The mask $d-1=2^r-3$ has all digits except digit one set, so Δx^{d-1} has $2^{r-2}-1$ odd terms. The right side has only the odd term x, forcing $r=3$. Here $\Delta x^5 = x^4+x+1$ and $\Delta x^4 = \Delta x^2 = 1$; of $b=2,4$, exactly $b=4$ satisfies the equation.

A missing position a in o corresponds to the two positions $2a+1$ and $n-2-2a$ in f. Thus $d=6$, $\{a,b\}=\{1,4\}$ gives $(14,3,4)$; $d=14$ and the three pairs from $\{2,4,8\}$ give the degree-30 cases in (17). The substitutions above also prove sufficiency. Reducibility follows uniformly from Proposition 3.7 with $s=2$. □

3.3 Full-degree quadratic constructions

Here a full-degree quadratic pullback over $\mathbb{F}_2$ means

$$F = h^m P(g/h), \qquad \gcd(g,h)=1, \qquad \max(\deg g, \deg h) = 2,$$

where $\deg P = m > 1$ and $\deg F = 2m$. The last condition is automatic for irreducible P; in general the degree drops exactly when $h_2 \neq 0$ and $P(g_2/h_2)=0$ [8, Section 2].

Corollary 3.9 (Boundary for full-degree quadratic pullbacks)**.** *If a full-degree quadratic pullback is a four-zero tail palindrome, then it belongs either to the lists in Theorem 3.3 or to* (17)*, and its degree is* 2 *modulo* 4*. If it is irreducible, it is either* $f_{14,2,4}$ *or a member of* (10)*. Consequently, no sequence of quadratic pullbacks, each doubling the degree, can reach a four-zero tail palindrome after starting at an even degree.*

Proof. Write $g=\sum_{j=0}^{2} g_j x^j$, $h=\sum_{j=0}^{2} h_j x^j$, and put

$$a = g_2h_1 + g_1h_2, \quad b = g_0h_2 + g_2h_0, \quad c = g_1h_0 + g_0h_1.$$

The classical quadratic framework gives $b^2+ac=1$ and, in the separable case,

$$(ax+b)^{2m} F\left(\frac{bx+c}{ax+b}\right) = F(x) \qquad \text{[8, Section 4, Lemma 9]}.$$

In the separable case the possible triples are exactly $(1,0,1),(0,1,1),(1,1,0)$, corresponding to reciprocity, translation and $\mathcal{M}_{2m}$. There is no nontrivial scalar normalization over $\mathbb{F}_2$. In the inseparable case $g_1=h_1=0$, the output is a square; its odd coefficients vanish, whereas a four-zero candidate has $[x^{2m-1}]F=1$.

The reciprocal branch contains no four-zero candidate: combining $f^*=f$ with Lemma 2.1 gives $(x+1)f = x^{n+1}+1 = (x+1)S_n$, hence $f=S_n$. The other two branches are classified by Theorems 3.3 and 3.8. The irreducible alternatives follow from their reducibility conclusions. All listed degrees are 2 modulo 4. Starting at an even degree, every subsequent doubled degree is divisible by four, proving the last assertion. □

The three involutions thus connect the support symmetries to an exhaustive list of full-degree quadratic outputs.

4 Root powers and coefficient reflection

Power substitution and taking powers of roots are different operations. For a monic $h \in \mathbb{F}_2[x]$ with constant term one, define

$$\mathcal{N}_\ell(h)(y) = \prod_{h(\alpha)=0} (y + \alpha^\ell), \tag{19}$$

where roots are counted with multiplicity. Its degree is $\deg h$, even when a root power has a shorter Frobenius orbit. It need not be the minimal polynomial of a single root power. This is the classical generalized Graeffe transform [9, Section 5], [10, Section 3.1]:

$$\mathcal{N}_\ell(h)(y) = \mathrm{Res}_z(h(z), y + z^\ell) = \det(yI + C_h^\ell),$$

where C_h is a companion matrix of h. These identities retain multiplicities, including for a reducible h. In contrast, $h(x^\ell)$ has degree $\ell \deg h$; classical order criteria for its irreducibility are recalled in [11, Section 2].

Theorem 4.1 (Cubic and quintic root-power images; computer-assisted)**.** *Let $f = f_{n,i,j}$ be a candidate from* (1)*. No irreducibility or squarefreeness assumption is imposed.*

1. *The polynomial $\mathcal{N}_3(f)$ is never tail-palindromic.*
2. *The polynomial $\mathcal{N}_5(f)$ is tail-palindromic if and only if*

$$(n,i,j) = (10,3,4) \quad \text{or} \quad (n,i,j) = (7d-1, 2d-1, 3d-1), \quad d \geq 1, \quad d \equiv \pm 1 \pmod{10}. \tag{20}$$

The first image has two missing tail coefficients; the second has exactly $2d$. In particular, neither root-power operation has a four-zero tail-palindromic image.

The finite parameter certificates used below are part of the proof. They cover all integer degrees and positions, rather than a bounded range of degrees. Appendix B specifies the exact support and integer-feasibility checks, with the supplied independent verification programs.

4.1 A sparse reflection defect

For odd ℓ and a primitive ℓth root of unity ζ,

$$\mathcal{N}_\ell(h)(x^\ell) = \prod_{s=0}^{\ell-1} h(\zeta^s x). \tag{21}$$

This follows by multiplying $x^\ell + \alpha^\ell$ over the roots of h. It also proves that the coefficients lie in $\mathbb{F}_2$. The operation is multiplicative, commutes with reciprocity, and sends $1 + x$ to $1 + y$. These are elementary norm identities; the object-specific issue is the coefficient reflection of their outputs.

Put $N = n + 1$, $\Delta = j - i$, $R = n - 1 - i - j$, and

$$\begin{aligned} D &= x^i(1 + x^\Delta)(1 + x^R), \qquad u = (1 + x)D, \\ P &= (1 + x)f = 1 + x^N + u, \qquad P^* = 1 + x^N + xu. \end{aligned} \tag{22}$$

The identity for P^* follows from Lemma 2.1. The eight raw terms of u may cancel when positions are adjacent; all such cancellations are retained. Thus P has at most ten terms, independently of n.

Let $g = \mathcal{N}_\ell(f)$, $Q_\ell = \mathcal{N}_\ell(P)$, and $F_\ell(x) = Q_\ell(x^\ell)$. Then $Q_\ell = (1+y)g$, so the exact tail-reflection test for g is

$$\begin{aligned}\Omega_\ell(x) &:= x^{\ell N}F_\ell(1/x) + x^\ell F_\ell(x) + (1+x^\ell)(1+x^{\ell N}) \\ &= (1+x^\ell)\big(g^*(x^\ell) + x^\ell g(x^\ell) + 1 + x^{\ell N}\big).\end{aligned} \tag{23}$$

Thus g is tail-palindromic if and only if $\Omega_\ell = 0$: the bracket vanishes precisely when its nonleading coefficients are a palindrome. In a homogeneous expansion in u, the highest-degree terms cancel between $\mathcal{N}_\ell(P^*)$ and $y\mathcal{N}_\ell(P)$. The remaining support still has bounded size for fixed ℓ.

Lemma 4.2 (Exact support certificates)**.** *For $k \geq 3$ and $1 \leq i < j < k$, the polynomial Ω_3 is nonzero. The polynomial Ω_5 is zero exactly at the parameter sets in* (20)*.*

Proof. Expand the fixed support of P in (22), separately in each residue class of (k, i, j) modulo ℓ. Use the positive gaps i, $j-i$ and $k-j$ to parametrize all allowed positions. For $\ell = 3$, the supplied certificate exhibits an uncancellable term in each of 42 parameter regions. For $\ell = 5$, the supplied collision cover has 672 nodes and leaves only the degree-10 exception and the two residue branches of the progression in (20). The construction, exhaustive coverage, and independent exact checks are detailed in Appendix B. Sufficiency for the two progression branches also follows directly from (24) below; the isolated exception is checked explicitly. These are finite parameter certificates, with no upper bound on the free integer parameters. □

4.2 The quintic progression and its zero count

Write $f_d = f_{7d-1,2d-1,3d-1}$, where $d \equiv \pm 1 \pmod{10}$, and put

$$\mathcal{A} = \{7, 8, 9, 12, 13, 14, 21, 22, 23, 26, 27, 28\}.$$

A compact support identity for the quintic progression is

$$\begin{aligned}(1+y)\mathcal{N}_5(f_d)(y) = 1 + y^{7d} + \sum_{a\in\mathcal{A}} y^{\lfloor ad/5\rfloor} \\ + \sum_{b=2}^{5}\big(y^{bd-1} + y^{bd}\big).\end{aligned} \tag{24}$$

One obtains this identity by splitting P into its five exponent residue classes and multiplying the five factors in (21). Equivalently, substitute $d = 10t+1$ or $10t+9$ into the 26-term norm form (35); the resulting affine exponent pairs are verified by the independent branch checker. This proves an identity in the free parameter t, including all cancellations at $d = 1$.

The support symmetry is visible in the formula. The set $\mathcal{A}$ is invariant under $a \mapsto 35 - a$, and

$$\lfloor ad/5\rfloor + \lfloor (35-a)d/5\rfloor = 7d - 1.$$

The terms $y^{bd-1} + y^{bd}$ are paired by $b \mapsto 7 - b$. Thus the right side of (24) satisfies the required reflection test.

For $d = 10t + \epsilon$, the 22 displayed exponents are distinct when $\epsilon = 1, t \geq 1$ or $\epsilon = 9, t \geq 0$. Order them as $0 = e_0 < e_1 < \cdots < e_{21} = 7d$. Division by $1+y$ makes the coefficient at degree s the parity of the number of exponents $e_r \leq s$. Hence the missing coefficients occupy the intervals $[e_1, e_2 - 1]$, $[e_3, e_4 - 1]$, and so on, and

$$\sum_{r=0}^{9}(e_{2r+2} - e_{2r+1}) = 20t + 2\epsilon = 2d. \tag{25}$$

The ordered affine supports and this sum are included in the branch certificate. At $d = 1$, cancellation gives

$$\mathcal{N}_5(f_{6,1,2}) = y^6 + y^5 + y^4 + y + 1,$$

again with two missing tail coefficients. The isolated case is

$$\mathcal{N}_5(f_{10,3,4}) = y^{10} + y^9 + y^8 + y^6 + y^5 + y^4 + y^3 + y + 1,$$

whose missing positions are 2 and 7.

Proof of Theorem 4.1. Equation (23) and Lemma 4.2 give the cubic obstruction and the complete quintic parameter list. Equations (24)–(25), together with the two explicit small images, give the asserted zero counts. Every allowed d is odd, so each quintic tail-palindromic image has 2 modulo 4 missing coefficients. It therefore cannot belong to the four-zero family. □

4.3 Irreducibility and degree-preserving extraction

Corollary 4.3. *For $\ell = 3$ or 5, there are no two irreducible four-zero candidates h, f of the same degree such that $h \mid f(x^\ell)$. If an irreducible four-zero polynomial has a root of order m, then*

$$\ell \not\equiv 2^r \pmod{m} \qquad (r \geq 0,\ \ell = 3, 5).$$

Proof. For a root β of h, the element β^ℓ is a root of f and has a full degree-n Frobenius orbit. Thus $\mathcal{N}_\ell(h) = f$, contrary to Theorem 4.1. If $\ell \equiv 2^r \pmod{m}$, taking the ℓth power merely permutes the conjugates, giving $\mathcal{N}_\ell(f) = f$ and the same contradiction. □

The infinite quintic shape branch itself is reducible except at $d = 1$:

$$f_d = S_{d-1}(x)\big(S_6(x^d) + x^{2d-1}(1 + x)(1 + x^{2d})\big). \tag{26}$$

Indeed $S_{7d-1} = S_{d-1}S_6(x^d)$ and the four deleted terms are $x^{2d-1}(1 + x^d)^3$. For $d > 1$ both factors in (26) are nonconstant. Since $\gcd(d, 5) = 1$, taking fifth powers permutes the nontrivial dth roots of unity, so $\mathcal{N}_5(S_{d-1}) = S_{d-1}$; the image is also reducible. Consequently the only irreducible inputs with a tail-palindromic quintic image are $f_{6,1,2}$ and $f_{10,3,4}$. Their root orders are respectively 63 and 341, as checked by irreducibility and exact-order tests in the supplied program. In particular, if $4 \mid n$, the quintic image of an irreducible four-zero candidate is not tail-palindromic.

Taking an ℓth root within $\mathbb{F}_{2^n}$ can increase a deficient ℓ-primary part of an element's order. Corollary 4.3 shows that for $\ell = 3, 5$ this degree-preserving one-step construction cannot also keep the four-zero tail. It does not classify arbitrary higher-degree rational transformations or constructions with intermediate polynomials outside the family.

5 Primitive counts and reproducible computations

Complete enumeration of the 161700 candidates at the 98 even degrees $6 \leq n \leq 200$ gives

$$\sum_{\substack{6 \leq n \leq 200 \\ n \text{ even}}} N_4(n) = 2663, \qquad N_4(n) > 0 \text{ at every tested degree.}$$

A separate implementation reproduces every complete count and first pair. At the 50 even degrees $202 \leq n \leq 300$, checking the lexicographic prefixes covers 5410 further candidates and verifies the first pair at each degree.

n	candidates	$N_4(n)$	first (i,j)
6	1	1	$(1,2)$
8	3	1	$(2,3)$
10	6	2	$(1,2)$
12	10	3	$(1,3)$
20	36	1	$(4,5)$
40	171	11	$(1,4)$
62	435	16	$(1,17)$
100	1176	31	$(1,23)$
126	1891	55	$(1,2)$
154	2850	45	$(1,23)$
194	4560	20	$(6,65)$
200	4851	17	$(1,71)$

Table 1: Selected complete counts. All 98 rows and all successful position pairs through degree 200 are supplied with the verification data.

Arithmetic verification and reproducibility. The verification code constructs coefficients directly and tests irreducibility, followed by the standard exact-order conditions

$$x^{2^n-1} \equiv 1 \pmod{f}, \qquad x^{(2^n-1)/p} \not\equiv 1 \pmod{f} \quad (p \mid 2^n-1 \text{ prime}). \tag{27}$$

The arithmetic tests use python-flint 0.9.0 with FLINT 3.6.0 [17]. The 148 order factorizations correspond to the 98 complete-count degrees and the 50 further first-witness degrees. Exact division leaves residual cofactor one; all 484 distinct prime factors, of at most 144 bits, pass FLINT's rigorous `fmpz_is_prime` procedure. A standard-library implementation independently agrees on all 220 candidates through degree 24.

The folder `verification/primitivity/` supplies complete candidate verdicts, successful positions, exact order-test residues for the 148 first witnesses, recovered integer factorizations and input hashes. Historical inputs and correction records are preserved with the audit. The data and verification programs are publicly archived at doi:10.5281/zenodo.23766722 and in the fixed GitHub release.

The arithmetic ranges and counts are unchanged in this revision. The new all-parameter proof certificates and standard-library checkers accompany the manuscript in `verification/root_power/`; see Appendix B. They form part of the root-power proofs and are supplied in addition to the earlier arithmetic records.

The companion programs also check the reciprocal filter, both quadratic classifications and the descent identities; detailed ranges accompany the archived outputs. The four translation outputs have independently certified factor degrees $2+12$, $12+18$, $6+24$ and $7+7+8+8$, with a repeated factor in the last. The B-sequence factorization data cover $3 \le R \le 10$: the first term is irreducible and the next seven reducible.

6 Structural consequences and open questions

The reciprocal screen counts positions compatible with reciprocal coprimality. The quadratic classifications determine every full-degree quadratic output and give a uniform obstruction from translation symmetry. The root-power classifications supply a different boundary: neither cubing nor taking fifth powers of all roots preserves the four-zero family.

Irreducibility in the remaining quadratic branch. Within full-degree quadratic constructions, the B family (10) remains the irreducibility question in Corollary 3.9. Which polynomials h_R of Proposition 3.5 are irreducible? The explicit descent and the binary Q criterion reduce this to classifying the irreducible outputs in that branch.

Primitive existence. Conjecture 1.1 still asks whether the prescribed support admits a primitive root at every even degree. The computations give finite evidence; no infinite-degree existence theorem is claimed. Corollary 4.3 excludes one-step degree-preserving cube- and fifth-root extraction as a construction within the family. It imposes necessary restrictions on root orders but does not force maximal order. The reducible quintic progression in (26) also shows why a nonempty infinite shape branch need not provide full-degree roots.

These are boundaries for specified constructions, not a nonexistence result for the conjecture. Higher-degree rational maps, selected factors, and constructions with intermediate polynomials outside the family require separate support and order arguments. The descriptions in Appendix A retain both conditions explicitly.

A Coefficient and finite-field descriptions

The coefficient quotient organizes position pairs into classes with explicit multiplicities; the finite-field parametrization encodes primitivity through coprime order components.

A.1 Coefficient quotient

Let ρ reverse $\mathbb{F}_2^k$, and set

$$P = \{(A, \rho A) : A \in \mathbb{F}_2^k\}, \quad K = \{(C, C) : C = \rho C\}, \quad U = P/K, \quad r = \lfloor k/2 \rfloor.$$

The map $[(A, \rho A)] \mapsto A + \rho A$ identifies U with $\mathrm{im}(I + \rho)$, of dimension r. Prepending a zero to $x \in \mathbb{F}_2^{k-1}$ defines $\Phi(x) = [(\widetilde{x}, \rho\widetilde{x})]$. The all-ones tail lies in K, so deleting positions i, j gives class $\Phi(e_i + e_j)$. With zero-based coordinates and a suitable basis $z_1, \ldots, z_r$, the columns $u_i = \Phi(e_i)$ form the multiset

$$\{u_1, \ldots, u_{k-1}\} = \begin{cases} \{z_1, z_1, \ldots, z_{r-1}, z_{r-1}, z_r\}, & k = 2r, \\ \{z_1, z_1, \ldots, z_{r-1}, z_{r-1}, z_r, 0\}, & k = 2r + 1. \end{cases} \tag{28}$$

Indeed pair positions $i, k - 1 - i$; the boundary pair contributes the singleton z_r and an unpaired middle position contributes zero. In particular Φ is onto.

For $s = \sum s_a z_a$, write $h = \sum_{a<r} s_a$ as an integer and $\varepsilon = k \bmod 2$. The ordinary coset weight enumerator is

$$A_s(z) = \sum_{\Phi(x)=s} z^{\mathrm{wt}(x)} = (1 + z)^{\varepsilon} z^{s_r} (2z)^h (1 + z^2)^{r-1-h}. \tag{29}$$

Each repeated column allows $00, 11$ for zero syndrome and $01, 10$ for nonzero syndrome; the singleton and optional zero column give the remaining factors. Thus $\mu(s) = [z^2]A_s(z)$ counts four-zero candidates in class s, and the number of occupied classes is

$$D(3) = 1, \qquad D(k) = 1 + \binom{\lfloor (k+1)/2 \rfloor}{2} \quad (k \geq 4). \tag{30}$$

This follows by counting the nonzero quadratic coefficients in (29); at $k = 3$ the zero class is unoccupied. For a character χ of U, put $t = \#\{i : \chi(u_i) = -1\}$ and $\ell = k - 1$. Then

$$\widehat{\mu}(\chi) = \sum_{i<j} \chi(u_i)\chi(u_j) = K_2(t; \ell) = \frac{(\ell - 2t)^2 - \ell}{2}. \tag{31}$$

These are classical direct-product weight-enumerator and Krawtchouk identities [7, Sections 3–4][6, Sections 1.1 and 2.1]. The different factorization behavior within a class is visible already at degree eight: $x^8+x^7+x^6+x+1$ is primitive, while $x^8+x^7+x^5+x^2+1 = (x^2+x+1)(x^6+x^4+x^2+x+1)$ is reducible; their tail difference 0110 0110 belongs to K.

A.2 Finite-field parametrization

Put $q = 2^k$, $\mathbb{K} = \mathbb{F}_q$, $\mathbb{L} = \mathbb{F}_{q^2}$ and $T = \{\gamma \in \mathbb{L}^* : \gamma^{q+1} = 1\}$. The coprime orders $q - 1, q + 1$ give $\mathbb{L}^* = \mathbb{K}^* \times T$. A primitive element is uniquely $\beta = \nu\gamma$ with $\mathrm{ord}(\nu) = q - 1$ and $\mathrm{ord}(\gamma) = q + 1$. Writing $\tau = \gamma + \gamma^{-1} \in \mathbb{K}$, the standard Dickson identity $D_a(u + u^{-1}) = u^a + u^{-a}$ and Frobenius pairing give

$$F_{\nu,\gamma}(x) = \prod_{i=0}^{k-1} \left(x^2 + \nu^{2^i} D_{2^i}(\tau)x + \nu^{2^{i+1}}\right). \tag{32}$$

Here $D_0 = 0, D_1 = t, D_{a+1} = tD_a + D_{a-1}$ in characteristic two [2]. Conversely, the two order conditions make this product the minimal polynomial of a primitive element. Expanding the product and comparing with (1) gives an exact coefficient-and-order system in which the support and the two exact-order conditions are imposed simultaneously.

B Exact parameter certificates for root powers

The computer-assisted part of Theorem 4.1 checks affine exponent collisions, not a finite list of degrees. This appendix specifies the complete input, the coverage argument and the arithmetic decision procedures. The accompanying `verification/root_power/` directory supplies the certificates and independent checkers as part of the proof.

B.1 Fixed support and parameter coverage

Represent the exponent $e_k k + e_i i + e_j j + e_0$ by (e_k, e_i, e_j, e_0). The complete input support of P in (22), before cancellations, is

$$\begin{gathered} (0,0,0,0),\ (2,0,0,1), \\ (0,1,0,0),\ (0,1,0,1),\ (0,0,1,0),\ (0,0,1,1), \\ (2,-1,0,-1),\ (2,-1,0,0),\ (2,0,-1,-1),\ (2,0,-1,0). \end{gathered} \tag{33}$$

In a fixed residue class $r = (k, i, j) \bmod \ell$, the coefficient of this term in $P(\zeta^s x)$ is

$$\zeta^{s(e_k r_k + e_i r_i + e_j r_j + e_0)}.$$

Multiplication, reciprocity and addition then give the full support of Ω_ℓ in (23). Every equality of specialized exponents is retained modulo two.

Set $a = i$, $b = j - i$, $c = k - j$. Let $a_0, b_0, c_0 \in \{1, \ldots, \ell\}$ be their positive residue representatives and write

$$a = a_0 + \ell A, \qquad b = b_0 + \ell B, \qquad c = c_0 + \ell C, \qquad A, B, C \in \mathbb{Z}_{\geq 0}.$$

This is a bijective parametrization of all admissible positions. An exponent in (33) becomes

$$\begin{aligned}\ell(e_k+e_i+e_j)A+\ell(e_k+e_j)B+\ell e_kC\\ +e_k(a_0+b_0+c_0)+e_ia_0+e_j(a_0+b_0)+e_0.\end{aligned}\tag{34}$$

A split into a variable equal to zero or at least one is disjoint and exhaustive; in the latter branch replace it by one plus a new nonnegative variable.

For $\ell=3$, splitting $P=p_0+p_1+p_2$ by exponent residues gives $F_3=p_0^3+p_1^3+p_2^3+p_0p_1p_2$ by (21). Direct expansion leaves 16–36 symbolic terms per residue class. The certificate covers the 27 residue classes by 42 regions: 41 are unbounded and one is the single point $(n,i,j)=(10,1,2)$. In each region a selected exponent w cannot equal any other remaining exponent v. Writing $d=w-v$, the equality would require

$$d_AA+d_BB+d_CC=-d_0,\qquad A,B,C\in\mathbb{Z}_{\geq 0}.$$

All 768 comparisons are refuted by 604 sign contradictions, 138 gcd nondivisibilities and 26 nonzero constants. Thus $\Omega_3\neq 0$ everywhere.

B.2 The quintic collision cover

The five-residue norm can be computed over $\mathbb{F}_2$. If $P=p_0+\cdots+p_4$ is split by exponent residues modulo five, the norm is the determinant of the circulant matrix $(p_{(j-i) \bmod 5})_{0\leq i,j<5}$. Its 26 terms have the compact expression

$$\begin{aligned}\Phi_5(p)=&\sum_{i=0}^{4}p_i^5+\prod_{i=0}^{4}p_i\\ &+\sum_{i=0}^{4}p_i^3(p_{i+1}p_{i-1}+p_{i+2}p_{i-2})\\ &+\sum_{i=0}^{4}(p_i^2p_{i+1}^2p_{i+3}+p_i^2p_{i+2}^2p_{i+1}),\end{aligned}\tag{35}$$

with indices modulo five. Diagonalizing the circulant over $\mathbb{F}_{16}$ gives the product in (21); expansion of the 120 determinant permutations independently gives (35).

The 125 residue classes are initially divided into 467 regions. Of these, 354 already have an uncancellable term and two have identically zero defect. The remaining 111 are resolved by the following exact collision procedure.

Under the accumulated affine equalities, first group identical exponents and cancel pairs. If the specialized polynomial is zero, any selected remaining term must have a partner of equal exponent. Branch over every feasible equality $E_w-E_v=0$. Each such branch raises the coefficient rank: an equality of unchanged rank would either have been cancelled already or be an inconsistent constant. With three variables, the rank can rise at most to three. The branches need only cover the zero set; they need not be disjoint. An accepting node certifies inclusion in (20); it need not independently prove zero defect. The explicit image formulas prove sufficiency separately. Induction on the increasing rank therefore proves necessity: each zero has a feasible partner at every nonterminal node, and a unique-term or infeasible node cannot contain a zero.

Nonnegative integer feasibility is decided exactly. At rank one, gcd divisibility is sufficient when coefficients have mixed signs: an integer solution can be shifted into the nonnegative orthant along

a strictly positive integral kernel vector. For coefficients of one sign, the target is either impossible by sign or belongs to a finitely checked additive semigroup. At rank two, Cramer's rule writes the variables as rational affine functions of one original integer coordinate t. Nonnegativity gives an interval, and the denominators give a finite integrality period; an interval-and-residue test is necessary and sufficient. The largest period in this certificate is three. At rank three, test the unique rational point for nonnegative integrality.

The resulting certificate has 672 nodes: 540 unique-term nodes, 127 collision nodes and five accepting nodes. All 28 611 candidate-partner feasibility checks are verified. The accepting sets are shown below; A, B, C refer to the initial gap variables, and a, b, c to the positive gaps.

(k, i, j) mod 5	conditions	resulting parameters
$(0, 3, 4)$	$A = B = C = 0$	$(10, 3, 4)$
$(3, 1, 2)$	$A = B = C = 0$	$d = 1$
$(1, 2, 1)$	$A, B \geq 1,\ C = 0;\ a = 2b - 1,\ b = 2c - 1$	$d = 9$
$(1, 2, 1)$	$A, B, C \geq 1;\ a = 2b - 1,\ b = 2c - 1$	$d = 10t + 9,\ t \geq 1$
$(3, 1, 2)$	$A, B, C \geq 1;\ a = 2b - 1,\ b = 2c - 1$	$d = 10t + 1,\ t \geq 1$

Putting $d = b$ in the last four rows gives $c = (d + 1)/2$, $a = 2d - 1$ and $n = 7d - 1$. The residue conditions give exactly $d \equiv \pm 1 \pmod{10}$. This proves the necessity part of Lemma 4.2.

B.3 Independent verification and supplied files

Run the following three programs in `verification/root_power/`:

```
python check_cubic_formal.py
python check_quintic_classification.py
python check_quintic_branches.py
```

They use only the Python standard library. The cubic checker reconstructs the norm from $a^3 + b^3 + c^3 + abc$, independently of the original three-factor $\mathbb{F}_4$ construction. The quintic classification checker uses the circulant determinant and Cramer parametrization, while the construction uses $\mathbb{F}_{16}$ products and Gaussian elimination. It checks every partition, every feasible collision partner and every accepting implication. The branch checker separately verifies (24), the ordered 22-term supports, the zero counts, the proper factors and the two small irreducible inputs. It also checks 250 actual polynomial substitutions across all residue classes; these are implementation controls, not the justification for an unbounded-degree conclusion.

The complete finite certificates in that directory are

```
cubic_certificate.json
quintic_classification_certificate.json
```

The certificates and programs constitute the finite part of the proof, with no SMT solver or upper degree cutoff required. An additional independent audit, `check_certificate_lattice.py`, reconstructs both norms in the quotient algebra $\mathbb{F}_2[y, z]/(z^\ell + y)$ and checks the same certificates using unimodular integer solution lattices. It imports neither the generators nor the three verification programs above.

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